CURRICULUM VITAE

PERSONAL INFORMATION

NAME:	Wookyoung Kim
DATE OF BIRTH:	September 27, 1993
NATIONALITY:	Korea
AFFILIATION:	Korea Institute of Machinery and Materials (KIMM), Heat Pump Research Center
POSITION:	Senior Researcher
CONTACT:	Heat Pump Research Center Korea Institute of Machinery and Materials (KIMM) 156 Gajeongbuk-ro, Yuseong-gu, Daejeon 34103, Korea (TEL) +82-42-868-7619 (E-MAIL) wookyoung@kimm.re.kr

EDUCATION

Degree	University	Department	Date
B.S.	KAIST	Department of Mechanical Engineering	February, 2015
M.S.	KAIST	Department of Mechanical Engineering	February, 2017
		Thesis: *"Effect of artificial cavities on the thermal performance of pulsating heat pipes"* **Advisor:** Sung Jin Kim	
Ph.D.	KAIST	Department of Mechanical Engineering	February, 2021
		Dissertation: *"Study on a Thermal Network-based Model for Predicting the Thermal Resistance of Pulsating Heat Pipes"* **Advisor:** Sung Jin Kim	

RESEARCH INTERESTS

(1) Data center cooling (immersion cooling, direct liquid cooling, jet-impingement cooling)
(2) High-heat-flux electronics cooling (CPU/GPU cold plate design)
(3) Heat exchangers (PCHE for cryogenic liquid hydrogen, compact heat exchanger design)
(4) Heat pump systems (chemisorption/adsorption heat pumps, high-temperature heat pumps)
(5) Low GWP refrigerant systems (next-gen eco-friendly refrigerants, VLE measurement)
(6) Two-phase heat transfer (boiling, evaporation, pulsating heat pipes, vapor chambers)
(7) Cryogenic heat transfer (liquid hydrogen vaporizer, supercritical fluid heat transfer)

PUBLICATIONS

SCI/SCIE Journal Papers

(1) B. Kim, K.H. Lee, M. Kim, S. Moon, J.W. Yoo, **W. Kim**, "Experimental and theoretical investigation of freezing phenomenon in Printed Circuit Heat Exchanger", *Applied Thermal Engineering* 273, 126455 (2025).

(2) D.H. Kim, **W. Kim**, J.H. Kim, H.S. Kim, K.H. Lee, "Experimental studies on the VLE of R-32/R-125 and verification of experimental setup", *Journal of Mechanical Science and Technology* 38(10), 5779-5785 (2024).

(3) H.S. Kim, D.H. Kim, J.S. Kim, **W. Kim**, Y. Kim, "A numerical study on the performance of chemisorption heat pump according to various design conditions", *Applied Thermal Engineering* 243, 122519 (2024).

(4) J.S. Kim, **W. Kim**, H.S. Kim, Y. Kim, "Pool boiling heat transfer of ammonia outside enhanced tubes with various fin structures", *Applied Thermal Engineering* 247, 122986 (2024).

(5) H.S. Kim, J.H. Kim, J.S. Kim, **W. Kim**, Y. Kim, "Experimental study of a chemisorption heat pump under different operation conditions", *Applied Thermal Engineering* 240, 122274 (2024).

(6) J. Kim, C.G. Kim, H.S. Kim, **W. Kim**, D.H. Kim, "Characterization of liquid behavior in distributor of falling-film evaporator", *Physics of Fluids* 35(6), 067124 (2023).

(7) J.S. Kim, **W. Kim**, H.S. Kim, Y. Kim, S.H. Yoon, "Pool boiling heat transfer of ammonia outside a single tube with fin structures: Hysteresis phenomena and boiling enhancement", *International Communications in Heat and Mass Transfer* 149, 107157 (2023).

(8) **W. Kim** and S.J. Kim, "Fundamental issues and technical problems about pulsating heat pipes", *Journal of Heat Transfer - Transactions of the ASME* 143, 100803 (2021).

(9) **W. Kim** and S.J. Kim, "Effect of a flow behavior on the thermal performance of closed-loop and closed-end pulsating heat pipes", *International Journal of Heat and Mass Transfer* 149, 119251 (2020).

(10) **W. Kim** and S.J. Kim, "Effect of reentrant cavities on the thermal performance of a pulsating heat pipe", *Applied Thermal Engineering* 133, 61-69 (2018).

KCI Journal Papers

(1) **W. Kim**, B. Kim, S. Sohn, K.H. Lee, J. Kim, "Experimental Investigation on the Freezing Condition of PCHE for Cryogenic Liquid Hydrogen Vaporizer", *Journal of Hydrogen and New Energy* 35(2), 240-248 (2024).

(2) **W. Kim**, H.S. Kim, J. Kim, D.H. Kim, "An Experimental Study on The Thermal Performance Measurement and Flow Visualization of Falling Film Evaporator Using R-1233ZD(E) Refrigerant", *Korean Journal of Air-Conditioning and Refrigeration Engineering* 36(1), 9-17 (2024).

(3) S. Sohn, **W. Kim**, "A Study on Anti-Icing Design by Conjugate Heat Transfer Analysis in a Lab-Scale PCHE for Supply of Cryogenic High Pressure Liquid Hydrogen", *Transactions of the Korean Hydrogen and New Energy Society* 33(5), 541-549 (2022).

(4) D.H. Kim, H.S. Kim, J. Kim, **W. Kim**, "An Experimental Study on the Performance Characteristics of a Falling Film Evaporator for R-1234ze(E) Refrigerant", *Korean Journal of Air-Conditioning and Refrigeration Engineering* 35(7), 363-370 (2023).

(5) J.S. Kim, S.A. Kim, D.H. Shin, **W. Kim**, S. Moon, Y. Chung, S. Sohn, "Comparative Study of Air Cooling and Immersion Cooling for the Thermal Management of a Cylindrical Battery Pack", *Transactions of the KSME B* 47(10), 543-550 (2023).

(6) H.S. Kim, **W. Kim**, K.H. Lee, D.H. Kim, "Numerical Study on the Performance of Hybrid Falling Film Evaporator", *Korean Journal of Air-Conditioning and Refrigeration Engineering* 34(8), 371-379 (2022).

(7) K.H. Lee, D.H. Kim, **W. Kim**, J.S. Kim, Y. Kim, J.W. Yoo, "Selection of the Equation of State for Thermodynamic Properties in the Low GWP Mixture Refrigerants", *Transactions of the KSME B* 46(10), 563-571 (2022).

International Conference Proceedings

(1) 2024, "Experimental and theoretical investigation on avoiding freezing phenomena of PCHE for cryogenic liquid hydrogen vaporizer", 29th ICEC

(2) 2024, "Understanding the thermal characteristics of falling film evaporation of R-1233ZD(E) refrigerant using flow visualization and heat transfer measurement", 11th ACRA

(3) 2023, "Experimental investigation on the flow and thermal characteristics of falling film evaporator using R-1233ZD(E) refrigerant", 11th ICBCHT

(4) 2019, "Comparison of thermal performance between CEPHP and CLPHP", 4th TFEC

(5) 2018, "Experimental investigation on the thermal performance of double-condenser pulsating heat pipes", 10th ICBCHT

(6) 2017, "Experimental investigation of artificial cavities on the thermal performance of a pulsating heat

pipe", InterPACK 2017

PATENTS

Domestic Patents (17)

(1) Heat exchanger with anti-freezing capability (1) **[2023.03.27]**

(2) Heat exchanger with anti-freezing capability (2) **[2023.03.27]**

(3) Modular fin-tube reactor **[2024.11.04]**

(4) Fin-tube type reactor **[2024.11.04]**

(5) Tray structure for heat exchanger **[2022.10.13]**

(6) Oil recovery system for refrigeration systems (1) **[2022.10.13]**

(7) Oil recovery system for refrigeration systems (2) **[2022.10.13]**

(8) Micro-channel reactor **[2023.04.19]**

(9) Immersion cooling device **[2024.10.25]**

(10) Geothermal heat supply device and heating system **[2023.10.17]**

(11) Heat exchanger thermal performance testing apparatus **[2023.06.16]**

(12) Ternary refrigerant composition and heat pump system **[2022.10.24]**

(13) Display device **[2023.10.20]**

(14) Duct assembly and combustor **[2022.03.30]**

(15) Immersion cooling HVAC system and method **[2022.12.22]**

(16) Heat pipe integrated reactor for adsorption heat pump **[2022.04.06]**

(17) Adsorption heat pump evaporator and system **[2023.03.31]**

SOFTWARE REGISTRATIONS

(1) PCHE thickness design program **(C-2024-052179)** [2024.12.11]

(2) High-temperature heat pump cycle design program **(C-2024-038691)** [2024.10.22]

(3) Heat exchanger HTC uncertainty analysis program **(C-2023-059071)** [2023.12.12]

(4) Lab-scale heat exchanger analysis program **(C-2023-056463)** [2023.11.30]

(5) 100 kg/hr hydrogen PCHE design program **(C-2025-056891)** [2025.12.11]

(6) Adsorption cooling simulation program **(C-2025-034652)** [2025.09.03]

(7) Vapor chamber performance analysis program **(C-2024-041200)** [2024.11.04]

(8) Fin-tube heat exchanger design program **(C-2024-038692)** [2024.10.22]

TECHNOLOGY TRANSFERS

(1) Heat pipe and vapor chamber structural design for electronics cooling

(2) Micro-channel bonding and fabrication techniques for heat exchangers

(3) Low dew-point dehumidifier development technology

(4) Ultra-thin vapor chamber for next-generation semiconductor cooling

(5) Compact inverter compressor-based dress care system development